

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
DBQuery	15	13	0.1193	0.0873	1.37 $\times$	0.1375	0.1664	0.83 $\times$

Table 2: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
DBQuery	0.2568	–	0.2537	–

Table 3: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
DBQuery	1.01 $\times$	–	–	–

Table 4: Runtime comparison of Gibbon (AoS/SoA), GHC, and MLton. Times are total median per iteration (s). **Bold** marks the fastest time for each program.

Program	Gibbon-AoS	Gibbon-SoA	GHC	MLton
DBQuery	0.2568	0.2537	–	–
Geomean	0.2568	0.2537	–	–

Table 5: Per-pass performance for `DBQuery`, ADT has 15 fields, SoA uses 13 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
clearQueryFlags	M	14/15	7%	0.0687	0.0830	0.83 $\times$
countJoins	F	3/15	80%	0.0186	0.0112	1.66$\times$
filterSelectivitySkew	F	5/15	67%	0.0183	0.0123	1.48$\times$
hashJoinPressure	F	5/15	67%	0.0199	0.0130	1.52$\times$
scaleCosts	M	15/15	0%	0.0688	0.0834	0.82 $\times$
sumCost	F	6/15	60%	0.0184	0.0149	1.24$\times$
sumMemory	F	6/15	60%	0.0185	0.0146	1.27$\times$
sumRows	F	6/15	60%	0.0255	0.0213	1.20$\times$
Total				0.2568	0.2537	1.01 $\times$
Geomean						1.22$\times$